

## Kaggle ML Explainability:

We want to explain **feature importance**

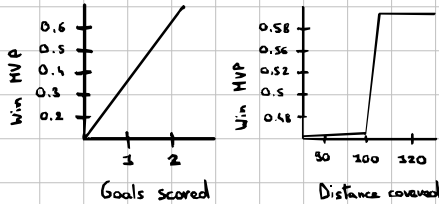
### **Permutation importance:**

- calculated AFTER a model is fitted
- randomly shuffles one column and observe new predictions
- ↳ `perm = PermutationImportance(model).fit(val_X, val_y)`  
`eli5.show_weights(perm, feature_names = val_X.columns.tolist())`

|                  | Weight          | Feature |
|------------------|-----------------|---------|
| More important ↑ | $0.2 \pm 0.02$  | Goals   |
|                  | $0.1 \pm 0.05$  | Saves   |
| Less important ↓ | $-0.5 \pm 0.02$ | Passes  |

### **Partial Plots:**

- calculated after a model is fit
- repeatedly alter the value for one variable (fix others)
- plot average predicted outcome



### **SHAP values:**

- tries to explain why a model made a prediction
- weighted average of feature's marginal contribution across coalitions

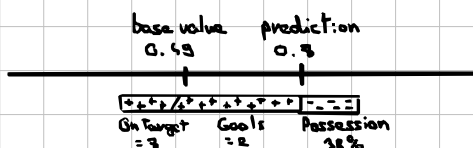
$$\text{Formula: } \phi_i = \sum_{S \subseteq F: i \in S} \underbrace{\frac{|S|! \cdot (|F| - |S| - 1)!}{|F|!}}_{\text{weights}} \underbrace{[f(S \cup \{i\}) - f(S)]}_{\text{diff. of predictions with/without feature } i}$$

Where:  $F$  = set of all features

$S$  = subset excluding feature  $i$

$f(S)$  = model prediction using  $S$

- $\sum (\text{SHAP values} \forall \text{ features}) = \text{pred\_for\_row} - \text{pred\_baseline\_value}$



: goal score biggest positive contribution